Borrow Research Conclusions

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Executive Summary

Combine (1) current borrow level, (2) v2 calibrated L2 stress score for spike risk, and (3) delta_borrow_5d_p50 OLS forecast for directional drift. Level/momentum features explain ~5% of forward borrow change; spike L0 remains too rare for calibrated probabilities.

We cannot reliably predict catastrophic borrow spikes (L0) with calibrated probabilities today. We can rank relative borrow stress (L2 tiers) and give a weak directional drift hint (~5% R2).

1. Three-Piece Borrow Outlook (Dashboard)

Piece 1: Level

Field: borrow_current. Spot annualized IBKR fee-only borrow. Strongest single anchor for future borrow.

Piece 2: Drift

Field: borrow_forecast_delta_5d_p50. Expected change in annual borrow over the next 5 borrow observations (git/IBKR history rows, not always 5 calendar days). Forecast level = borrow_current + delta. Pooled OLS on borrow_current, borrow_slope5, borrow_vol10, borrow_z60. Use directionally only; low R2 (~5%).

Piece 3: Stress

Field: borrow_spike_alert_tier (watch / elevated / high). Based on L2 calibrated probability. Use tier names for monitoring, not literal L0 spike percentages.

2. Drift Forecast Definition

Target: delta_borrow_5 = borrow at t+5 observations minus borrow at t (decimal annual rate).

Model: delta_borrow_5 ~ intercept + b1*borrow_current + b2*borrow_slope5 + + b3*borrow_vol10 + b4*borrow_z60, fit pooled across all symbols in borrow_predictor_panel.parquet.

- Positive delta: borrow likely rising; short carry may worsen vs spot.
- Negative delta: borrow easing; spot may overstate long-run cost.
- Does NOT change net_edge_p50 or spike tier logic.

3. L2 Calibrated Tier (Deep Summary)

L2 label (training target)

spike_event = 1 if BOTH: (a) max borrow over next 5 observations > 90th percentile of last 60 obs, and (b) jump > 10 percentage points annual. ~841 positives in replay vs 1 for L0 catastrophic.

Model pipeline

- logistic_v2: L2-regularized logistic regression on borrow dynamics + supply/scale features.

- Isotonic calibration: bin raw scores, map to realized L2 rate, enforce monotonicity.
- Alert tiers on calibrated p: watch >= 5%, elevated >= 12%, high >= 25%.

L2 replay metrics

- Rows: 35454 | Positives: 846
- AUROC: 0.777665 | ECE: 0.006819
- elevated: avg pred 0.146, realized 0.129 (n=1694)
- high: avg pred 0.501, realized 0.242 (n=207)
- low: avg pred 0.013, realized 0.017 (n=33553)

L0 catastrophic (not for sizing)

- Replay positives: 1 in 29869 rows. Elevated/high L0 bands showed 0% realized catastrophic rate. Do not use P(spike) >= 30%.

4. Key Features Explained

borrow_vol10

Std dev of day-over-day borrow changes over last 10 observations. Measures borrow choppiness, not level. Rank-corr ~0.12 with 5-obs delta borrow.

log_aum

log(1 + AUM) from etf_metrics_daily (NAV x shares_outstanding fallback). Fund size proxy. Rank-corr ~0.11 with delta borrow at h=5.

shares_available

IBKR-reported shares available to borrow. Drives shares_drop*, utilization_proxy, near_zero_shares. Rank-corr ~0.10 with delta borrow. Broker-specific and noisy.

5. Net Edge and Borrow (Unchanged)

net_edge_p50_annual still comes from ls-algo: block-bootstrap gross drag, inverse-variance blend with forward Exp. edge, minus resampled borrow history (weighted_empirical, 90d halflife). Anchor: borrow_for_net_annual ~ borrow_posterior_annual.

Borrow outlook is annotation only: drift and stress tiers do not modify net edge math. YieldBOOST borrow-once invariant preserved.

6. Research Programs

Program A - Spike predictor accuracy

Walk-forward replay, live scoring, borrow_spike_eval.json metrics. Dual labels L0 + L2.

Program B - Borrow predictor study

Panel: 46009 rows, 503 symbols. Best ablation: borrow_plus_supply. Supply helps slightly; peer basket hurts holdout (excluded from v2).

- Top rank-corr predictors at h=5: borrow_current, borrow_vol10, log_aum.

7. Milestone Status

- [BLOCKED] Enough L0 catastrophic spikes for stable eval: 1 / 20
- [DONE] L2 relative-stress label positives for model dev gates: 846 / 100
- [IN_PROGRESS] Daily prediction archive depth: 59 / 60
- [DONE] Supply feature coverage for logistic_v2: 0.9958 / 0.8
- [DONE] Calibrated elevated tier realized rate > 0: 0.12928 / 0.05
- [IN_PROGRESS] logistic_v2 L2 AUROC >= v1 L0 AUROC on replay: 0 / 1
- [DONE] Boosting L2 passes precision@10 lift gate vs base rate: 1 / 1
- [IN_PROGRESS] Rolling 60d elevated/high tier L2 realized rate: 0.0 / 0.05

Next actions

- Unblock l0_replay_positives_20: 1 vs target 20

8. Recommended Next Steps

- Fix utilization coverage (42% -> 80%): improve shares_outstanding joins in metrics panel.
- Keep daily L0 prediction archive; do not productize L0 probabilities until >= 20 replay positives.
- Optional: net_edge_stress_p50 display column using max(posterior, forecast) for elevated tier.
- Optional: live calibration monitor on elevated-tier hit rate over rolling 60d.
- Do NOT: peer features in v2, replace ls-algo borrow with drift forecast, L0 prob sizing.

9. Artifacts and Commands

- data/borrow_spike_tracking.json - milestone tracker
- data/borrow_spike_eval.json - replay metrics L0 + L2
- data/borrow_forecast_latest.json - per-symbol drift
- data/borrow_predictor_study_summary.json - feature rankings
- BORROW_SPIKE_EVAL.md - operator guide

Run: python scripts/borrow_spike_pipeline.py

Refresh dashboard rows: python scripts/build_data.py --borrow-only